THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL MCQ TEST SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Medium level: multi-step questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. Two vertical poles of heights 40 m and 22 m stand 24 m apart on level ground. The distance between their tops (in m) is: 1

(a) 42 **(b)** 28 **(c)** 30 **(d)** 31

2. A sum of ₹ 1200 buys 24 kg of rice at ₹ 50 per kg. When the price falls to ₹ 40 per kg, the same money buys (kg): 1

(a) 30 **(b)** 24 **(c)** 48 **(d)** 28

3. A polyhedron has 12 faces and 30 edges. Its number of vertices is: 1

(a) 20 **(b)** 18 **(c)** 40 **(d)** 44

4. The mean of 5 numbers is 28. The number to be added so that the mean becomes 30 is: 1

(a) 45 **(b)** 30 **(c)** 2 **(d)** 40

5. A rectangle has length $(3x + 2)$ cm and breadth $(2x + 5)$ cm. Its perimeter (cm) is: 1

(a) $10x + 7$ **(b)** $6x^2 + 10$ **(c)** $10x + 14$ **(d)** $5x + 7$

6. A path 2 m wide runs around the outside of a 20 m by 15 m rectangular garden. The area of the path is: 1

(a) 16 m² **(b)** 140 m² **(c)** 456 m² **(d)** 156 m²

7. $\sqrt[3]{0.027}$ is equal to: 1

(a) 0.4 **(b)** 0.03 **(c)** 0.3 **(d)** 3.0

8. $2^3 \times 5^3$ equals: 1

(a) 133 **(b)** 1000 **(c)** 250 **(d)** 30

9. The number of digits in the binary form of 20 is: 1

(a) 6 **(b)** 4 **(c)** 7 **(d)** 5

10. In a kite ABCD with AB = AD and CB = CD, $\angle A = 60^\circ$ and $\angle C = 130^\circ$. The angle $\angle B$ is: 1

(a) 170° **(b)** 75° **(c)** 85° **(d)** 95°

11. The smallest number to be added to 4829 to make it divisible by 8 is: 1

(a) 5 **(b)** 6 **(c)** 4 **(d)** 3

12. The value of $(x + 2)(x + 5)$ when $x = 4$ is: 1

(a) 58 **(b)** 26 **(c)** 54 **(d)** 44

13. A : B = 2 : 3 and B : C = 6 : 5. If A + B + C = 165, then C equals: 1

(a) 55 **(b)** 33 **(c)** 66 **(d)** 44

14. A number is increased by 25% and the result is 300. The original number is: 1

(a) 480 **(b)** 216 **(c)** 264 **(d)** 240

15. The area of a right triangle with hypotenuse 30 cm and one leg 18 cm (in cm²) is: 1

(a) 432 **(b)** 270 **(c)** 234 **(d)** 216

16. A train at 60 km/h takes 9 hours for a journey. At 45 km/h it will take (hours): 1

(a) 13 **(b)** 11 **(c)** 12 **(d)** 6.75

17. A cube of side 4 cm is painted all over and cut into 1 cm cubes. The number of cubes with no face painted is: 1

(a) 4 **(b)** 24 **(c)** 56 **(d)** 8

18. The mean of the first 9 multiples of 4 (that is 4, 8, ..., 36) is: 1

(a) 20 **(b)** 18 **(c)** 22 **(d)** 40

19. If $x + y = 9$ and $xy = 14$, then $x^2 + y^2$ equals: 1

(a) 81 **(b)** 109 **(c)** 67 **(d)** 53

20. A right triangle has hypotenuse 25 cm and one leg 15 cm. Its area is: 1

(a) 187 cm² **(b)** 250 cm² **(c)** 150 cm² **(d)** 300 cm²

21. The value of 45^2 is: 1

(a) 2050 **(b)** 2030 **(c)** 2025 **(d)** 2000

22. The product $(2 \times 10^3) \times (3 \times 10^4)$ in scientific notation is: 1

(a) 6×10^7 **(b)** 6×10^6 **(c)** 60×10^7 **(d)** 6×10^8

23. $1011_2 + 110_2$ equals: 1

(a) 10000_2 **(b)** 10011_2 **(c)** 10001_2 **(d)** 10010_2

24. Three exterior angles of a quadrilateral are $80°$, $100°$ and $95°$. The fourth exterior angle is: 1

(a) $75°$ **(b)** $95°$ **(c)** $110°$ **(d)** $85°$

25. The smallest 4-digit number divisible by 7 is: 1

(a) 1001 **(b)** 994 **(c)** 1008 **(d)** 1000

26. $(x + 2)^2 - (x - 2)^2$ equals: 1

(a) $4x$ **(b)** $2x^2 + 8$ **(c)** $8x$ **(d)** $8x + 4$

27. On a map with scale 1 : 50000, a distance of 4 cm represents: 1

(a) 20 km **(b)** 200 km **(c)** 6 km **(d)** 2 km

28. The difference between CI and SI on ₹ 5000 for 2 years at 4% p.a. (in ₹) is: 1

(a) 9 **(b)** 16 **(c)** 7 **(d)** 8

29. Which set of numbers is a Pythagorean triplet? 1

(a) 34, 44, 55 **(b)** 33, 44, 55 **(c)** 33, 46, 55 **(d)** 33, 44, 56

30. Two numbers are in the ratio 7:11 and their sum is 144. Their difference is: 1

(a) 72 **(b)** 32 **(c)** 18 **(d)** 34

31. A Koch curve starts with a line of length 27 cm. The total length after stage 2 (in cm) is: 1

(a) 15.19 **(b)** 54 **(c)** 72 **(d)** 48

32. On a line graph, a town's population (in thousands) rises from 20 to 30. The percentage increase is: 1

(a) 25% **(b)** 60% **(c)** 10% **(d)** 50%

33. The sum of 3 consecutive even numbers starting with $2n$ is: 1

(a) $6n + 8$ **(b)** $3n + 6$ **(c)** $6n + 3$ **(d)** $6n + 6$

34. The diagonal of a square is 10 cm. Its area is: 1

(a) 100 cm² **(b)** 50 cm² **(c)** 20 cm² **(d)** 25 cm²

35. The smallest number by which 98 must be multiplied to get a perfect square is: 1

(a) 2 **(b)** 1 **(c)** 3 **(d)** 4

36. The value of $2^9 - 9^2$ is: 1

(a) 431 **(b)** 593 **(c)** 0 **(d)** -431

37. In 2403_5, the value (in decimal) contributed by the digit 4 in position 2 from the left is: 1

(a) 20 **(b)** 500 **(c)** 100 **(d)** 29

38. A quadrilateral whose diagonals bisect each other and are equal is a: 1

(a) kite **(b)** rectangle **(c)** trapezium **(d)** rhombus

39. The value of 83 - 38 is: 1

(a) 36 **(b)** 135 **(c)** 54 **(d)** 45

40. Simplifying $5(x + 2) - 3(x - 4)$ gives: 1

(a) $2x - 2$ **(b)** $8x + 22$ **(c)** $2x + 22$ **(d)** $2x + 14$

41. 36 km/h expressed in m/s is: 1

(a) 6 m/s **(b)** 10 m/s **(c)** 12 m/s **(d)** 129.6 m/s

42. Two successive discounts of 10% and 10% are given on a marked price ₹ 2000. The selling price (in ₹) is: 1

(a) 1600 **(b)** 1782 **(c)** 1458 **(d)** 1620

43. An isosceles triangle has equal sides 39 cm and base 30 cm. Its height (in cm) is: 1

(a) 37 **(b)** 36 **(c)** 24 **(d)** 35

44. In a pie chart of 144 students a sector has a central angle of 80°. The number of students it represents is: 1

(a) 8 **(b)** 32 **(c)** 64 **(d)** 35

45. The total number of triangles removed from the Sierpinski triangle up to stage 3 is: 1

(a) 13 **(b)** 26 **(c)** 9 **(d)** 27

46. The mean mark of 20 students is 35 and of another 30 students is 45. The mean of all 50 students is: 1

(a) 41 **(b)** 43 **(c)** 40 **(d)** 2050

47. On expanding $5(x + 2) + 4x$ we get: 1

(a) $20x + 10$ **(b)** $9x + 2$ **(c)** $9x + 10$ **(d)** $5x + 14$

48. An L-shaped figure is made by removing a 10 cm by 7 cm rectangle from a 25 cm by 18 cm rectangle. Its area is: 1

(a) 450 cm² **(b)** 380 cm² **(c)** 165 cm² **(d)** 520 cm²

49. The smallest number by which 250 must be divided to get a perfect cube is: 1

(a) 3 **(b)** 2 **(c)** 1 **(d)** 4

50. $2^{-1} + 3^{-1}$ equals: 1

(a) $\frac{1}{5}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{5}{6}$ **(d)** $\frac{2}{5}$

END OF QUESTION PAPER **Total: 50 Marks**

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